Predictors of Adolescent Additional-Language Attainment: A Systematic Review and Meta-analysis

Lois Ann Parri[1], Tomos Jones Parry[2], Sara Les[1], Justine Kwek[1], Brechtje Post[3], Mirjana Bozic[1]

[1]*Department of Psychology, University of Cambridge;* [2]*School of Modern Languages, The Open University;* [3]*Department of Theoretical and Applied Linguistics, University of Cambridge.*

Brain, Language & Bilingualism Laboratory

UNIVERSITY OF CAMBRIDGE

1. Background

Adolescents vary substantially in how successfully they acquire an additional language, even when they experience broadly similar instructional opportunities [1]. Attainment variability has been linked to various learner differences, including cognitive abilities [2], affective factors [3], prior linguistic experience [4], and opportunities for language exposure [5].

As well as predictors, there are numerous outcomes for additional-language attainment (overall proficiency; broad language skills; or finer linguistic features), which are closely related, but not interchangeable.

See for yourself...

Look at the posters here today: how many ways can you spot attainment being measured and reported?

The existing evidence spans numerous disciplines: cognitive neuroscience, education, psychology, etc. This also partly contributes to the differing methods for defining and measuring the same predictors and outcomes.

However, the fragmented evidence base and the pooling of finer features may obscure meaningful predictor-outcome relationships. A predictor strongly associated with one aspect of language learning may show a weaker or different relationship with another. This specificity disappears as outcomes merge.

Aim: To synthesise the fragmented evidence base and examine how learner and environmental predictors are associated with adolescent additional-language outcomes at progressively finer levels.

2. Methods

Eligibility: This pre-registered [6] review included quantitative studies with adolescents aged 10-24 years learning an additional language in instructional settings, that report an association between a learner-level or environmental predictor and a language-attainment outcome. Self-reported proficiency and experimentally manipulated predictors were excluded.

Searching and Screening: The searches were conducted on 19 November 2025 and updated on 20 August 2026 across PubMed, MEDLINE, Embase, Scopus, Web of Science, and ProQuest Dissertations and Theses Global, and supplemented by hand-searching. Records were screened independently by two reviewers for both title/abstract and full-text. Reviewers calibrated after screening 10% and resolved conflicts at the end of each stage.

Data Extraction: Record, sample, predictor, outcome, and effect-level data were extracted. Risk of bias was assessed using the appropriate Joanna Briggs Institute Critical Appraisal Checklist [7].

Outcome Hierarchy: Language outcomes were categorised into nested and progressively finer levels.

Level 1) global proficiency

Level 2) language skills
reading; writing; listening; speaking

Level 3) linguistic features
lexical semantics, morphosyntax; syntax; phonology; orthography; pragmatics; morphology.

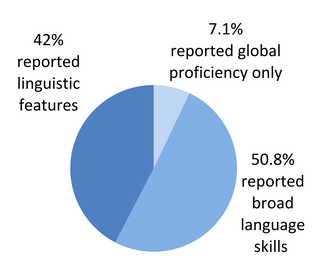
Synthesis and Meta-Analysis:

1) Descriptive synthesis and evidence-density mapping.
2) Random-effects meta-analysis for associations meeting the minimum evidence threshold following Cochrane guidance [8] (5 records, 10 effects).
3) Multilevel meta-regression to test outcome category as a moderator.

3. Results

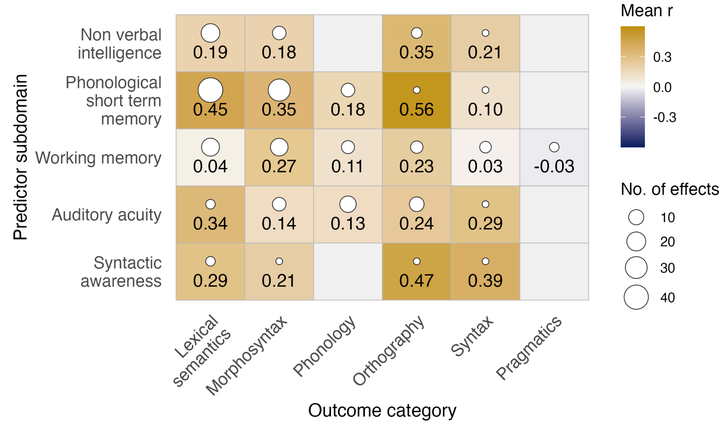
Level 1) Global Proficiency: **175** records; **3,558** effects; **137,720** participants

Of the 3,558 effects in the overall model, 1,798 (50.8%) were classified at the language-skill level, 1,508 (42%) at the linguistic-feature level, and 252 (7.1%) at the global-proficiency level. Skill labels included reading (575 effects), writing (468), speaking (414), listening (323), and combinations of these (18). Linguistic-feature effects comprised lexical semantics (724), morphosyntax (354), syntax (194), phonology (130), orthography (74), pragmatics (18), and morphology (14). The overall model showed high heterogeneity ($I^2 \approx 86\%$), suggesting substantial variation at the finer predictor-outcome levels. For feasible reporting, this poster focuses on cognitive predictors; click/scan the QR code for more results.



Level 2) Language Skills: Cognitive predictors were positively associated with all four skills: reading (r = .35), writing (r = .34), speaking (r = .27), and listening (r = .24). However, these associations were not uniform: the skill-level moderator test was significant (p < .001), and Holm-corrected comparisons showed stronger associations for reading and writing than for speaking and listening. Other predictors showed a more skill-specific pattern, e.g., instruction quantity was most strongly associated with reading (r = .36, p = .002), compared with listening (r = .21, p = .059), writing (r = .18, p = .228), and speaking (r = .16, p = .117).

Level 3) Linguistic Features: At the finest level, the heatmap shows that cognitive predictor associations varied across linguistic features. E.g., phonological short-term memory had stronger descriptive mean associations with lexical semantics (r = .45) and morphosyntax (r = .35), whereas working memory showed a weaker mean association with lexical semantics (r = .04) than with morphosyntax (r = .27). The moderator test was not significant (p = .203), so the differences should be treated as descriptive. Many finer predictor-by-feature cells were sparse; an unestimated cell indicates limited evidence, not evidence of no association.



4. Discussion

Global outcomes provide a useful broad summary, but they may obscure differences in how learner characteristics relate to particular language skills. Skill- and feature-specific findings may inform the future development and evaluation of adaptive materials, and give educators a fuller picture of learners' strengths and needs. This research could contribute to more responsive learning experiences for young people.

The hierarchical synthesis was a strength, allowing broad trends to be compared with skill- and feature-specific patterns. However, evidence was uneven and sparse across finer associations, limiting firm conclusions and potentially reducing the ability to detect feature-level moderation.

These results highlight the need for well-powered studies that use consistent measures and detailed reporting, particularly for underrepresented outcomes, e.g., linguistic features, to explore whether potential feature-specific patterns are reliable.

SCAN/CLICK TO DISCOVER MORE

References

1. Tagarelli et al., 2016. 2. Turker et al., 2021. 3. Botes et al., 2022. 4. Hirosh & Degani, 2018. 5. Muñoz & Cadierno, 2021. 6. Parri et al., 2025. 7. JBI Critical appraisal Tools. 8. Cochrane Handbook, 2024. See full references by clicking/scanning the QR code.